

Introduction to Animal Eye Anatomy

The eye structure divides into three main parts: accessory structures around the eye, the eyeball, and refractive media chambers. Accessory structures include the orbit (bony cavity housing the eyeball, muscles, nerves, and tear-producing/draining components), located laterally in herbivores like cattle for panoramic vision. The eyeball comprises three concentric coats: fibrous (sclera and cornea), vascular pigmented (uvea: choroid, ciliary body, iris), and nervous (retina). Eye chambers are vitreous (largest, posterior), anterior (between cornea and iris), and posterior (small, between iris and lens); refractive media include cornea, aqueous humor, lens, and vitreous body.

Vision in animals like cattle enables survival functions such as finding food and detecting predators, converting light into electrochemical impulses via the retina and optic nerve to the brain. Domestic mammals' visual systems adapt to varied lighting, with herbivores emphasizing wide-field views. The document covers general domestic animal anatomy with relevance to cattle, noting species variations like lateral orbits.

Accessory Structures

Orbit and Periorbita

The orbit is a bony socket formed by lacrimal bone (nasolacrimal duct fossa), maxilla (infraorbital canal), temporal bone, and pterygopalatine fossa; caudally continuous. In cattle, goats, and horses, orbits position laterally for panoramic vision, unlike forward-facing in predators. Contains eyeball, extraocular muscles, nerves, vessels, and lacrimal apparatus. Periorbita (orbital fascia) separates eyeball from orbital fat; apex attaches at optic foramen, base to orbital bones.

Eyelids (Palpebrae)

Three eyelids: upper, lower, and third (nictitating membrane at medial canthus). Upper/lower meet at lateral (small, round) and medial (larger, U/V-shaped) canthi; medial forms lacrimal lake (recess) with caruncle (lymph nodules, well-developed in equines). Palpebral fissure is elliptical opening; cilia (short, curved hairs) more on upper lid edges in cattle, sheep, goats. Layers from outside: skin (thin, pliable), muscular (orbicularis oculi innervated by facial n. VII; **levator palpebrae** superioris by oculomotor n. III; **corrugator supercilii** and levator anguli oculi

Levator palpebrae

From Latin levator **L**ifter, one that raises **R** and palpebrae from palpebra **E**yelid **L**iterally **L**ifter of the eyelid **R**efers to the muscle that raises the upper eyelid.

Corrugator supercilii

Latin corrugare **C**rinkle, draw into folds **R** and supercilium **E**yebrow **L**iterally **C**rinkler of the eyebrow **R**edescribes the small facial muscle that draws the eyebrows together and produces vertical forehead lines.

medialis by facial n.), fibrous tarsus (embeds meibomian glands), palpebral conjunctiva.

Glands: Moll's (modified sweat, near cilia), Zeis (sebaceous in follicles), tarsal/meibomian (secrete phospholipids for tear film stability, reduce evaporation).

- Functions: Protection, tear distribution.

Conjunctiva and Third Eyelid

Conjunctiva: thin mucous membrane with palpebral (lines eyelids) and bulbar (lines sclera) parts; fornices at reflections. Third eyelid: mobile, glandular at medial inferior sac; consists of nictitating membrane (semilunar fold), T-shaped hyaline cartilage (wide superficial, narrow deep, concave ventral), 12-16 excretory ducts in dorsal fornix, lymph glands (superficial around cartilage mid, deep Harderian in ruminants/pigs). Well-developed in camels/birds for dust protection; ill-developed in most domestics.

Lacrimal Apparatus

Produces/drains tears to moisten cornea, reduce friction. Components:

- Lacrimal gland: dorsolateral orbit, two surfaces (dorsal convex to frontal bone fossa, ventral deep); secretes tears.
- **Puncta lacrimalis**: dorsal/ventral slits near medial lid edge (one in rabbit/pig).
- Lacrimal canaliculi: converge to lacrimal sac.
- Lacrimal sac: dilated nasolacrimal duct origin in lacrimal bone fossa.
- Nasolacrimal duct: 25-30 cm rostroventral from medial canthus to nasal vestibule floor; parts: osseous canal, nasolacrimal groove (cartilage/mucosa-covered), terminal opening at middle meatus-skin junction.

Tears circulate cornea to lake, enter puncta/canaliculi.

Extraocular Muscles and Fat

Six recti/obliques (dorsal/ventral rectus, medial/lateral rectus, dorsal oblique tendon over cartilage, ventral oblique), retractor bulbi, levator palpebrae superioris. Orbital fat cushions, separated by periorbita.

Eyeball (Bulbus Oculi)

Fibrous Tunic

Outermost protective: cornea (anterior, transparent, avascular, refracts light) and sclera (posterior, opaque, fibrous). Cornea: convex anterior, concave posterior, thickest peripherally; layers: epithelium, Bowmans capsule, stroma, Descemet's membrane (joins iris via pectinate ligament), endothelium. Nerve supply: ophthalmic n. V. Sclera: collagenous, covered by bulbar conjunctiva/episclera (loose collagen/elastic, vessels, melanocytes), stroma, lamina fusca (to choroid);

Puncta lacrimalis

Latin punctum  point, small spot  (plural puncta) and lacrimalis from lacrima  tear  Literally  little tear points  Denotes the tiny openings on the eyelid margins that drain tears.

external surface for muscle insertion, internal to choroid. Limbus (corneoscleral junction): groove with venous plexus draining anterior chamber aqueous humor.

Vascular Tunic (Uvea)

Middle pigmented/vascular layer nourishes avascular parts, reduces light scatter (melanin).

- Choroid: posterior 5/6, vascular for oxygenation/nutrition; anterior via ciliary body.
- Ciliary body: triangular, ciliary muscle (accommodation via lens shape change), processes secrete aqueous humor (outflow via Schlemm's canal).
- Iris: anterior ciliary extension, covers lens anteriorly, controls pupil diameter (pigment determines color); thin circular structure.

Nervous Tunic (Retina)

Innermost, cup-shaped double layer: outer pigmented (to choroid), inner neural (to vitreous). Parts: pars optica (posterior 4/5, both layers, to ora serrata/ciliaris), pars ceca (anterior 1/5, pigmented only: **pars iridica** at iris, **pars ciliaris** at ciliary body).

Responsible for image perception.

Eye Chambers and Refractive Media

Chambers

- Anterior: cornea endothelium to iris anterior epithelium; aqueous humor, communicates with posterior via pupil.
- Posterior: narrow, iris posterior to lens/suspensory ligaments; aqueous humor.
- Vitreous: lens anterior to retina posterior; vitreous body (largest).

Refractive Media

Light refracts through: cornea (transparent barrier), aqueous humor (anterior/posterior chambers), lens (biconvex, cellular: anterior epithelium, posterior fibers in cortex/nucleus; bathed anteriorly by posterior aqueous, posteriorly by vitreous; shape-changes for focus),

Pars iridica

Latin pars **part** and iridica from iris **rainbow**, colored ring of the eye. Literally **iris part**. Refers to the portion of a structure (such as the retina or ciliary body) associated with the iris.

Pars ciliaris

Latin pars **part** and ciliaris from cilium **eyelash**; edge of the eyelid. Later extended to **ciliary body**. Literally **ciliary part**. Indicates the segment related to the ciliary body of the eye.

vitreous humor (fills chamber).

Vitreous humor

Latin vitreus **glassy**, made of glass **and** humor **moisture, fluid**. Literally **glassy fluid**. Describes the clear, gel **like** substance filling the space behind the lens of the eye.